S & B Design Limited

Building Management System

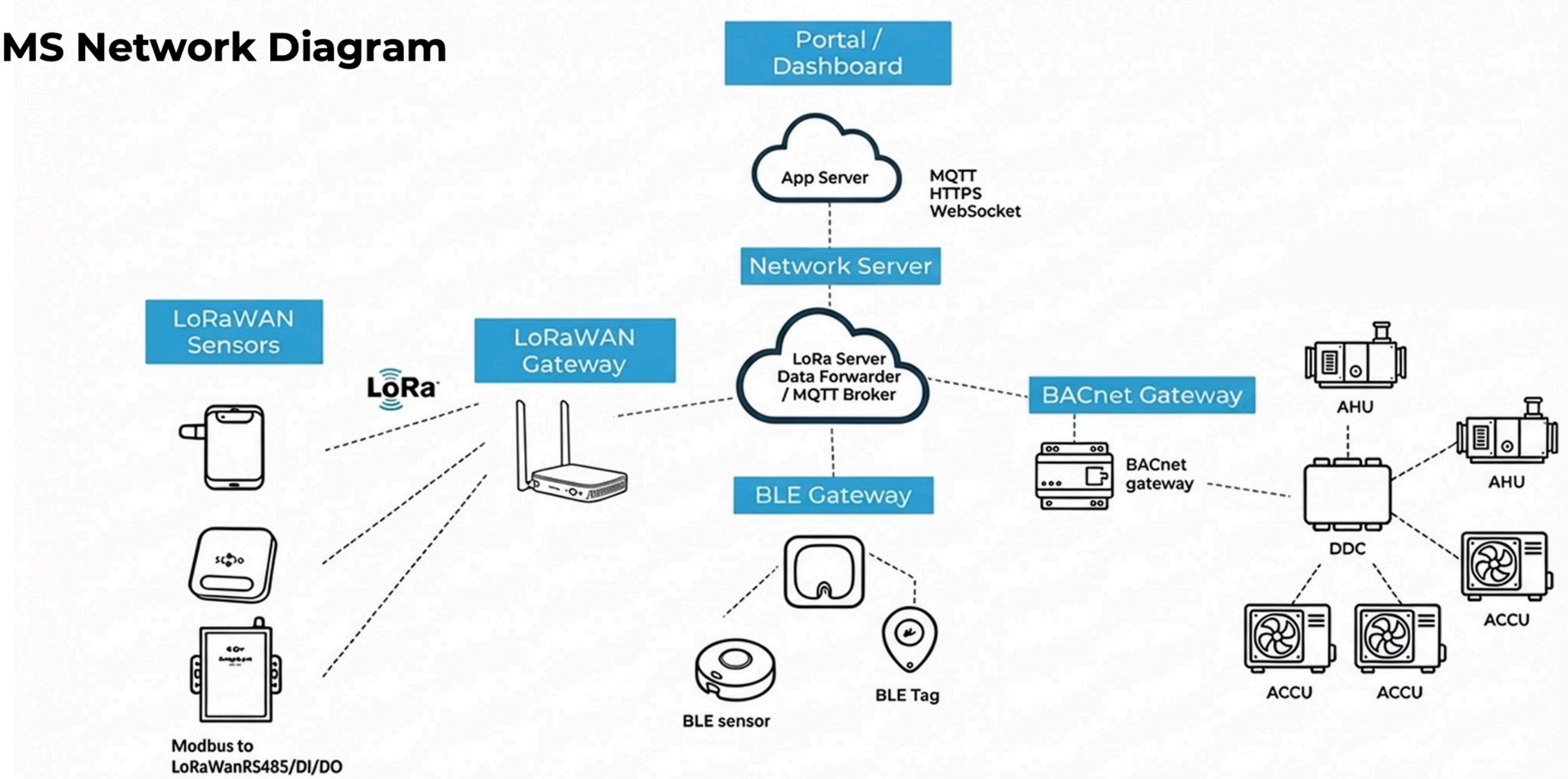


Hybrid BMS with wired and wireless technologies

Our Building Management System (BMS) is a true hybrid system that intelligently combines wired protocols (Modbus and BACnet) with wireless LoRaWAN® technology in a single, unified platform.

- Wired protocols deliver rock-solid, low-latency performance for core and critical equipment
- LoRaWAN® wireless adds flexible, low-power sensors and devices for extended coverage, easy expansion, and cost-effective retrofitting

Hybrid BMS Network Diagram



Key Benefits of Hybrid BMS



Lower installation & retrofit costs
No extensive new cabling, trenching, or disruption in existing/occupied buildings.



Battery-powered efficiency
LoRaWAN devices last 4 – 10+ years on batteries → minimal maintenance.



Wider & faster coverage
LoRaWAN penetrates walls/floors and reaches remote plant rooms, rooftops, or multi-story/campus areas easily.



Richer data & better optimization
More granular sensing enables advanced energy savings, fault detection, and occupant comfort (often 20–40% energy reduction potential).



Scalability & flexibility
Add hundreds of new sensors quickly and cheaply without rewiring.



Seamless upgrade path
Keeps your existing Modbus/BACnet investments while adding modern IoT capabilities.



Reliability with redundancy
Wired ensures stable control for mission-critical loops; wireless provides monitoring backup and failover.

This hybrid approach is especially practical for retrofits, smart buildings, campuses, or facilities in dense urban, where full rewiring is expensive or disruptive.

Centralized Monitoring & Control

WSmart centralizes your entire IoT ecosystem, providing the tools you need to monitor, manage, and act with confidence. Our platform is fully responsive, delivering a seamless experience on desktop, tablet, and mobile.



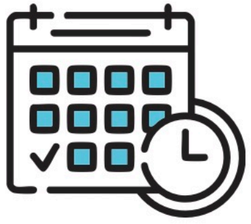
Intuitive Dashboard
Gain a real-time, at-a-glance overview of your entire device fleet's health and status.



Advanced Alert Handling
Receive instant, actionable notifications by SMS/WhatsApp/Email to minimize downtime and resolve issues proactively.



Robust User Management
Assign granular permissions and roles to ensure secure access and efficient team collaboration.



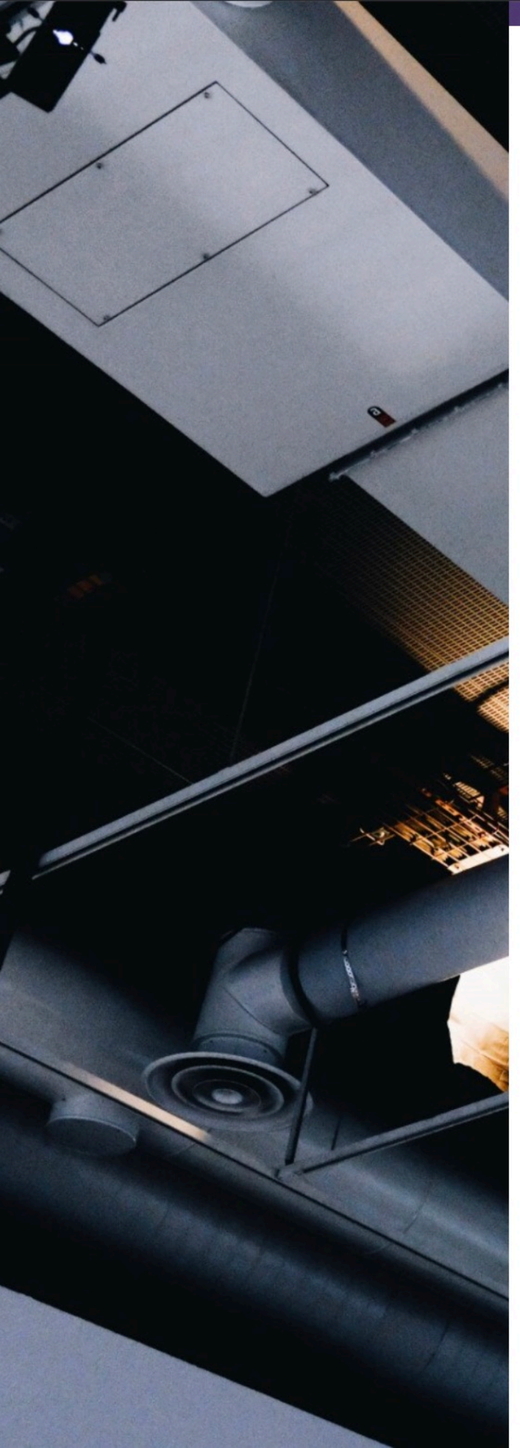
Scheduling
Schedules automate the operation of various building systems, such as HVAC, lighting, and security, based on pre-set parameters.



Report
Generate reports based on specific needs and timeframes, enabling trend analysis and forecasting.

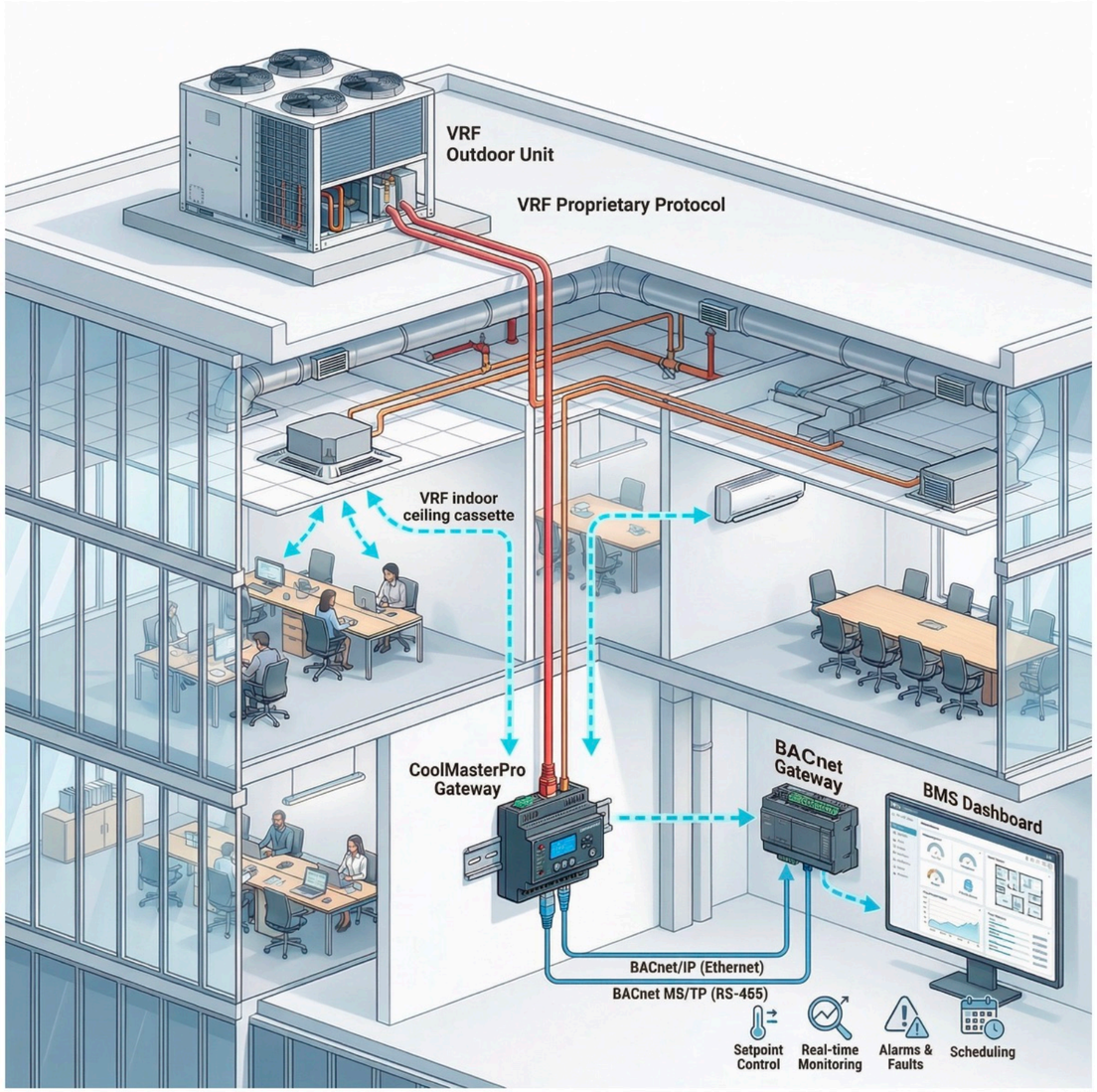


Accessible Anywhere
With a mobile-first design, you have the full power of WSmart in your pocket, 24/7.



HVAC (Heating, Ventilation, Air Conditioning) Management

CoolMasterPro serves as a universal gateway that seamlessly integrates VRF HVAC systems (Daikin, Mitsubishi, LG, etc.) with Building Management Systems (BMS) over BACnet. It connects directly to the VRF proprietary communication bus, auto-discovers indoor units, and exposes full control (on/off, mode, setpoint, fan speed) plus diagnostics data as standard BACnet objects. This enables centralized monitoring, control, alarming, and energy optimization from any BACnet-compatible DDC or BMS platform – plug-and-play for medium-to-large commercial installations.



CoolMaster

- Control and management VRF HVAC system
- Communication protocol - bacnet/mobus
- Scheduling control for individual unit
- Remote ON/OFF control
- MAX CONNECTED UNITS: 255



HVAC - Additional Sensors for Enhancing Building IAQ & Hygiene with LoRaWAN Sensors

Complementing CoolMasterPro VRF control, install LoRaWAN-connected Air Quality Sensors, Odor Detectors (in toilets), and Vape Detectors to elevate indoor environmental quality. These sensors integrate seamlessly into the BMS for real-time monitoring and automated responses (e.g., ventilation boost, alerts, cleaning dispatch).

Key Benefits of Installing These Sensors:



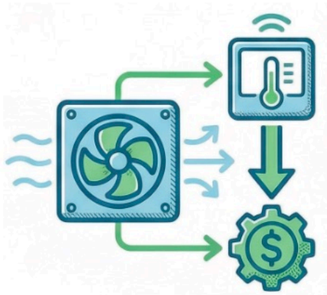
Improved Occupant Health & Comfort
Real-time detection of pollutants (CO_2, VOCs, particulates), odors (ammonia/H_2S in restrooms), and vape aerosols reduces respiratory risks, allergies, and exposure to harmful chemicals.



Policy Enforcement & Safety
Vape detectors deter unauthorized vaping in offices/schools, protect air quality, ensure compliance, and support wellness initiatives.

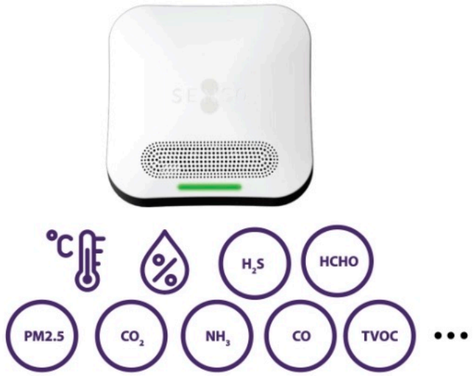


Better Hygiene & User Experience
Odor detectors trigger proactive cleaning in toilets, minimizing complaints and maintaining pleasant facilities.



Energy Optimization
Data enables demand-controlled ventilation, reducing unnecessary HVAC runtime and energy costs.

HVAC - Additional Sensors for Enhancing Building IAQ & Hygiene with LoRaWAN Sensors



Wireless indoor air quality sensor

- Extreme long battery life, up to 5 years
- Types of monitoring: **Temperature, Humidity, PM2.5, CO2, TVOC**
- And can detect any two from below gases to a sensor: **CO, NO2, NH3, H2S, HCHO, O2, O3, H2, ETO, SO2, NO, PH3, CLO2, CL2, HCL, HCN, HC3SH, HF, C2H2, C2H4**



Vape Detector

- Multi-Sensor Integration
- Real-Time Status Alerts
- Resilient Against Environmental Interference
- Adjustable Buzzer Hibernation Duration
- Instant Flame Alarming
- Anti-Tamper & Theft Protection



Odor Detector

- Detect NH_3 & H_2S with Higher Accuracy
- Built-in Buzzer & Traffic Light Status Indicator for Alarming
- Up to 3 Years Battery Life



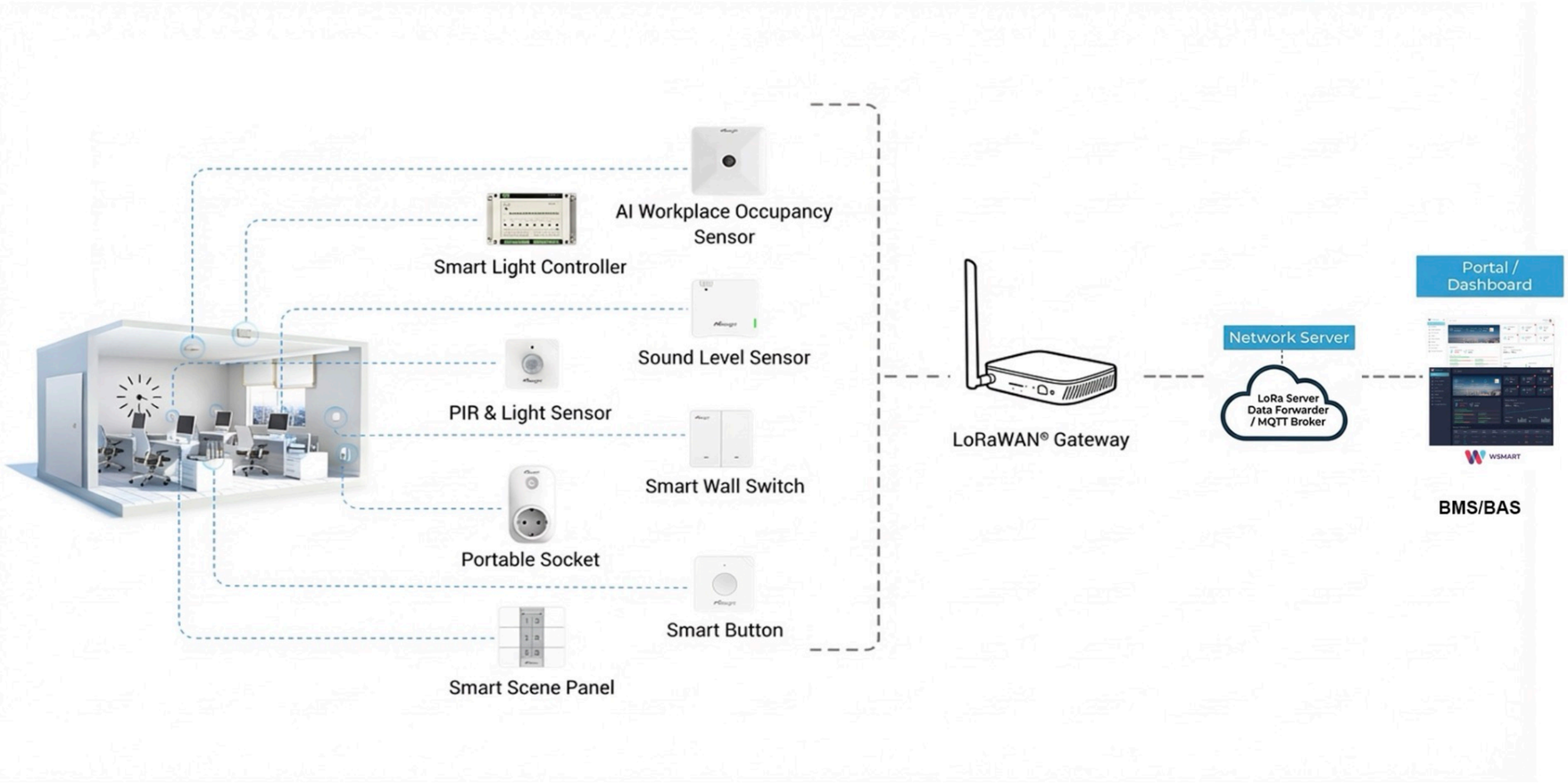
Ambient Temp/Humidity Sen sor

- Temp range: -20 to 60C
- Up to 5 years battery life
- Over/under threshold Alert



Lighting Control

This intelligent lighting control system integrates the Smart Light Controller, Smart Wall Switch, and PIR & Light Sensor via LoRaWAN® for seamless BMS integration. The Smart Light Controller centrally manages up to 8 lighting circuits with power monitoring, while wall switches provide local override and PIR/light sensors enable automatic occupancy-based control. All devices connect wirelessly to the BMS for remote monitoring, scheduling, and optimization.



Lighting Control

Key Benefits of Installing These Sensors:



Energy Savings
Occupancy detection (PIR) + ambient light sensing auto-off lights in unoccupied areas; power metering helps identify waste → up to 30–50% lighting energy reduction.



Convenience & User Comfort
Manual local control via wall switches + automatic on/off based on presence/light levels → no manual intervention needed.



Easy Retrofit & Low Installation Cost
Fully wireless LoRaWAN deployment → no extensive rewiring; quick setup in existing buildings with minimal disruption.



Centralized BMS Management
Real-time status, scheduling, remote control, and integration with other systems (e.g., HVAC) for unified building automation.



Reliability & Low Maintenance
Long-range wireless coverage, battery-powered sensors (5–10+ years life), overcurrent protection on controller.



Scalability
Easily add more devices across floors/rooms for granular control in offices, schools, or commercial spaces.

Lighting Control



Smart Light Controller

- Control up to 8 Lighting Circuits
- Remote & Time-Based ON/OFF
- Power Consumption Measurement
- Power-Off Memory Function



Smart Wall Switch

- Remote ON/OFF at Your Command
- Track Power Consumption with Precision



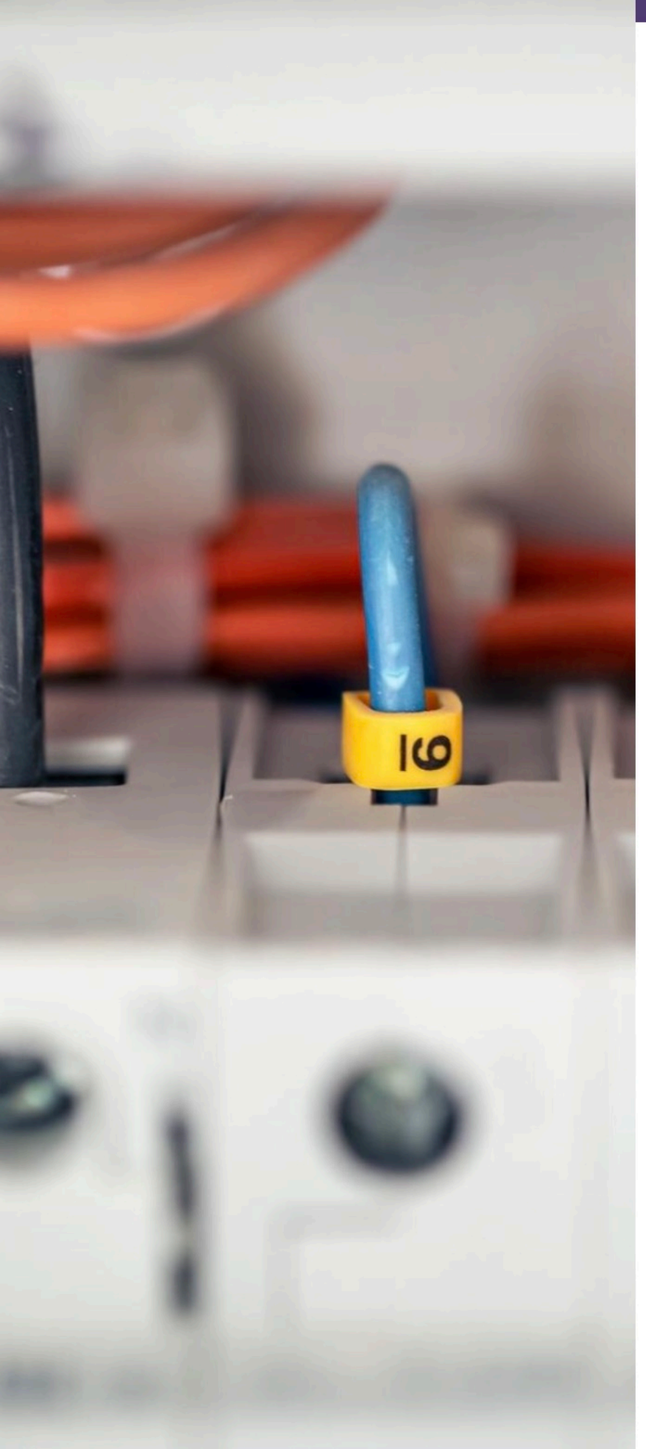
Smart Scene Panel

- Adapt to Multiple Scenarios
- Customize Display for Specific Needs
- Efficient One-to-Many Control



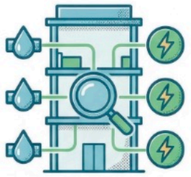
PIR & Light Sensor

- Experience Precise PIR Presence Detection
- Accurate Brightness Measurement



Energy Management & Optimization

Real-time monitoring and optimization of energy and water usage across the building through granular sub-metering per floor or area. LoRaWAN power meters and water meters provide accurate consumption data, seamlessly integrated into the BMS for visualization, trend analysis, anomaly detection, and automated efficiency actions.



Granular visibility of electricity and water consumption per floor / zone



Identify inefficiencies, leaks, and high-usage areas quickly



Achieve significant cost savings through targeted reduction measures



Enable demand-based control and predictive alerts



Fully wireless LoRaWAN installation – fast, low-cost retrofit, minimal disruption



Long battery life (5–10+ years) with low maintenance



Centralized dashboards, alarms, and ESG/compliance reporting

Energy Management & Optimization



Power Meter

- Monitor the power with real time data showing in IoT portal/dashboard
- Support 100A / 200A / 400A / 600A CT
- NB-IoT Wireless Connectivity
- Up to 4 x 3 phase or 12 x Single phase AC metering
- Real-time measurement: Voltage, Current, Power, Power Factor, Frequency, Energy



Current Meter

- Clamp-on current transformer
- SX1276 wireless communication module
- 2 ER14505 battery (3.6V/section) in parallel power supply
- Main body: IP53; sensor: IP30
- Magnetic base
- LoRaWAN Class A compatible
- Frequency Hopping Spread Spectrum (FHSS)
- Improved power management for longer battery life



RS485/Modbus Bridge to LoRaWAN

- LoRaWAN 1.0.3 protocol stack, supports Remote cloud management of RS485 devices
- Can address up to 16 RS485, a queue with up to 32 instruction sets
- Industrial grade STM MCU: ultra-low-power (4uA sleep) and wide temperature range of operation.
- Can power RS485 devices via a dedicated output.



MIU for Water Meter

- Compatible interface with Itron Aquadis+ piston type water meter
- Bidirectional flow measurement
- Leakage, backflow, peak flow, MIU tamper detection
- Battery operation (>15 years)
- IP68



People Counting & Occupancy Monitoring

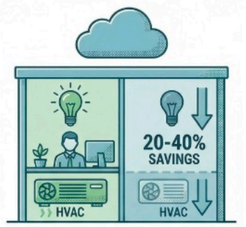
Enhance building intelligence, space utilization, and user experience with accurate, privacy-friendly people counting and occupancy detection integrated into the BMS. Using AI ToF People Counting Sensors for offices and exhibition centers, and Toilet Occupancy Sensors for restrooms, the system delivers real-time headcount, zone occupancy status, and usage insights. Data flows via LoRaWAN® (or wired protocols) to the BMS for centralized visualization, analytics, automated responses, and optimized space management.



Optimized Space Utilization
Real-time occupancy data identifies underused or overcrowded areas in offices and exhibition halls → enables better space planning, hot-desking, desk booking, and layout redesign



Enhanced Visitor Experience
In exhibition centers, people counting supports crowd flow analysis, queue management, and dynamic signage for better attendee satisfaction



Energy & HVAC Savings
Demand-controlled ventilation and lighting based on actual presence → reduce energy waste in unoccupied zones (typically 20–40% savings on HVAC/lighting)



Decisions
Centralized BMS dashboards show trends, peak hours, utilization rates, and reports to support staffing, space allocation, and sustainability goals

People Counting & Occupancy Monitoring



AI Stereo Vision People Counter

- 2nd Generation ToF, Anonymous, GDPR-compliant, AI-powered.
- Bi-Directional/Regional People Counting, Dwell Time Detection



PIR & Light Sensor

- PIR detection and brightness measurement, allowing it to determine an indoor area's occupancy or vacancy status effortlessly.
- Experience Precise PIR Presence Detection
- Accurate Brightness Measurement



AI Workplace Occupancy Sensor

- LoRaWAN® & PoE Versions for Broader Applications
- Up to 95% Industry-Leading Accuracy
- 100% Anonymous Detection (Compliant with GDPR)
- Smart People Flow Analysis
- Smart U-Turn Detection
- Ultra Large Detection Area
- Maximum 16 Detection Regions



Sound Level Sensor

- 30 to 120 dBA Measurement Range
- ± 3 dB Accuracy; 0.1 dB Resolution
- Weighting, C-Weighting & Fast-Time Weighting
- Measurement Unit: SPL Actual, Leq, and Lmax

Toilet Occupancy Monitoring

The smart restroom management system uses integrated sensor
improve hygiene, efficiency, and user experience in office and cor
buildings.



Toilet Occupancy Sensor

- Detects real-time usage of each stall/urinal for accurate occupancy status.
  Dual Technology of PIR and ToF
- Industry-leading Accuracy exceeding 99.5%
- Compliant with GDPR



Smart Fill Level Sensor

- Monitors toilet paper (and optionally soap/hand towel) levels to prevent shortages. Suitable for Diverse Toilet Paper Dispenser
- Accurate Fill Level Monitoring with ToF Technology
- Up to 2 Years Battery Life with One Coin Cell Battery



Odor Sensor

- Measures air quality (ammonia, H_2S,) to detect poor odor levels and trigger ventilation or cleaning alerts
- Built-in Buzzer & Traffic Light Status Indicator for Alarming
- Up to 3 Years Battery Life





Urban Garden management

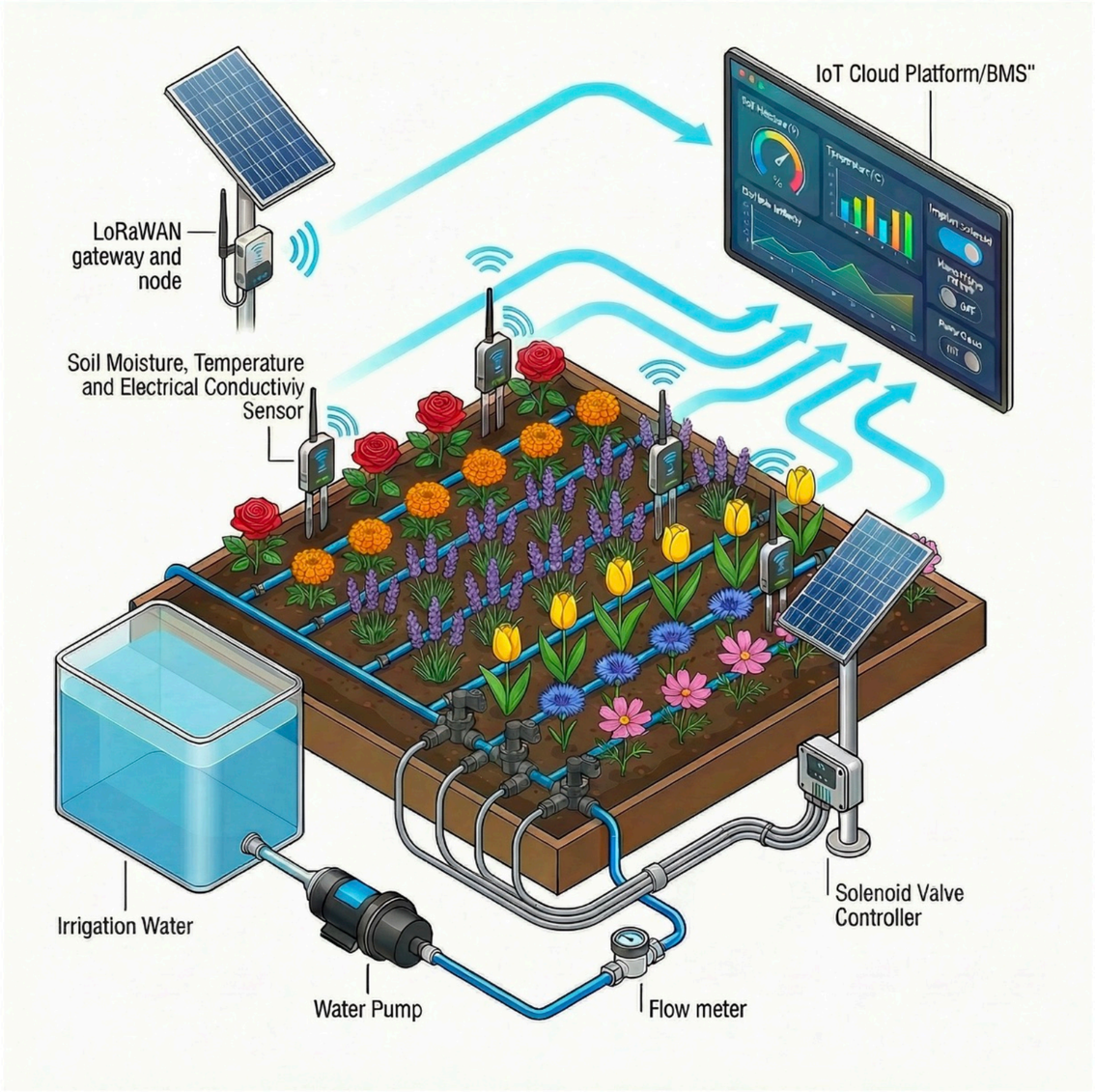
This smart irrigation solution leverages LoRaWAN® wireless technology for precise, automated, and efficient water management in agriculture, landscaping, or green spaces.

LoRaWAN Solenoid Valve Controller
Remotely controls DC latching solenoid valves for on/off and precise irrigation timing/volume.

LoRaWAN Soil Moisture & Temperature Sensor
Monitors soil conditions in real-time to trigger irrigation only when needed.

LoRaWAN Flow Meter
Measures water usage accurately via pulse counting for volume-based control and leak detection.

Solenoid Valve
Executes commands from the controller to regulate water flow to drippers or sprinklers.



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Urban Garden management

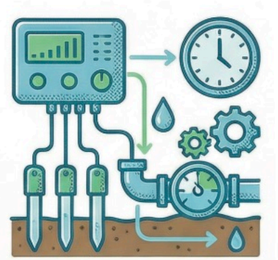
Key Benefits:



Water Savings
Up to 30–50% reduction by irrigating only when soil moisture is low, avoiding overwatering.



Remote Management
Monitor & control from anywhere via cloud dashboard; receive alerts for anomalies or leaks.



Precision & Automation
Real-time soil data + flow monitoring enables smart, demand-driven irrigation without manual intervention.



Improved Crop/Plant Health
Optimal moisture & temperature prevent stress, boost yields, and support sustainable practices.



Cost Efficiency
Low-power LoRaWAN reduces wiring/installation costs; long battery/solar life minimizes maintenance.

Urban Garden management



Solenoid Valve Controller

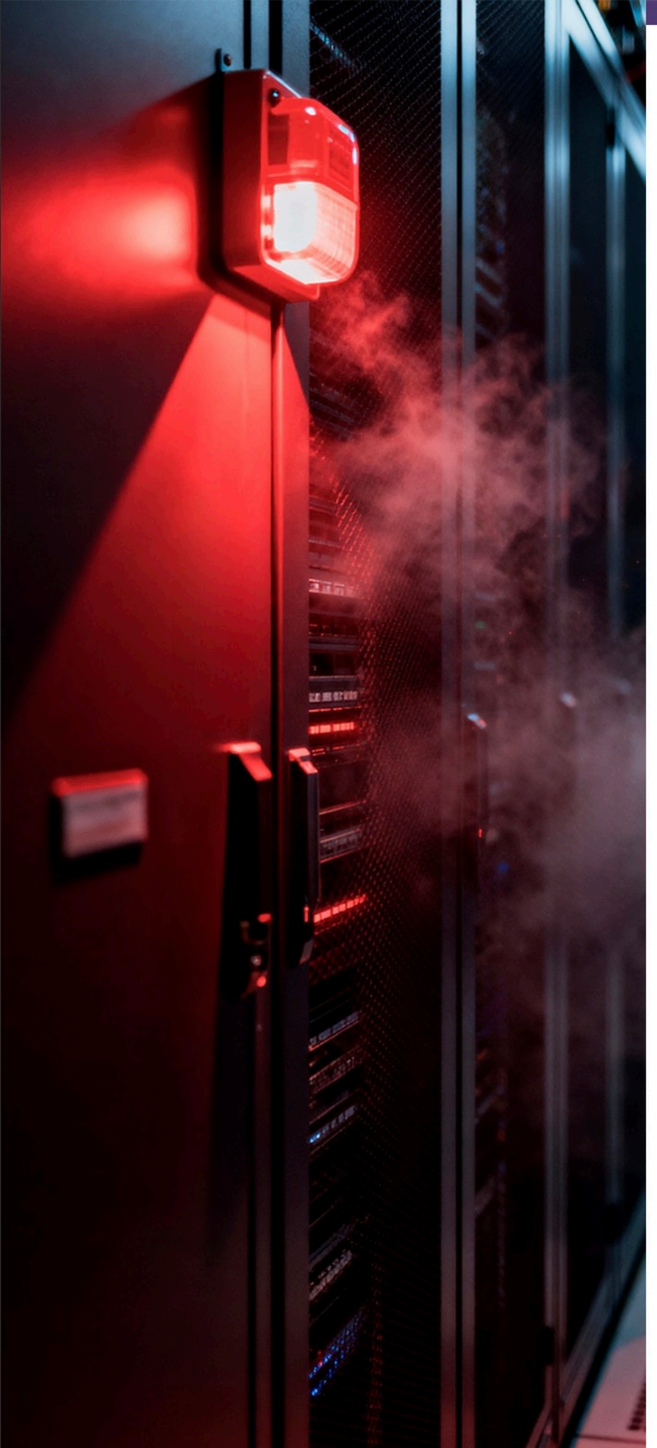
- Solenoid Valve Control
- Autonomous Operation
- Flexible Power Supply Options
- Support Multicast for Control in Bulk
- IP68 Rated Enclosure for Harsh Environment
- Internal and External Antenna Versions Optional
- Data Integrity and Reliability



Soil Moisture, Temperature, and Electrical Conductivity Sensor

- Seamless Integration with Smart Irrigation Systems
- IP67 Rated Transceiver for Harsh Environment
- IP68 Rated Corrosion Resistance Probe
- Data Integrity and Easy Management
- Long-Lasting Battery Life of up to 10 Years

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Critical Case Management

Critical case management in the BMS focuses on early detection and rapid response to high-risk events that can cause property damage, safety hazards, business disruption, or health risks. The system integrates dedicated LoRaWAN® sensors for water leakage, extreme temperature, and smoke detection. These sensors provide real-time alerts to the BMS, triggering automated actions (e.g., shut-off valves, HVAC shutdown, notifications to security/facility teams) and logging events for post-incident analysis.



- Analog level measurement
- Up to 5 years battery life
- Leakage and Cable Break Alert

Water Leakage Sensor
(e.g., placed under AHUs, piping, server rooms, plant rooms)

Use Case: Detects water leaks from pipes, condensate drains, or equipment failures.

BMS Action: Immediate alert + automatic solenoid valve closure to prevent flooding; notify maintenance team; log event for insurance claims.

Benefit: Minimizes water damage, mold risk, and costly downtime (especially critical in data centers, offices, and basements).

Critical Case Management



- High precision PT100 Temperature probe range from -200˚C to 600˚C
- Alert when: higher/lower than pre-set temperature threshold
- Up to 3-year battery life
- Easy installation without any complex and pricey set up cost

Extreme Temperature Sensor
(high/low threshold monitoring in server rooms, electrical panels, plant rooms, cold storage)

Use Case: Monitors for overheating (e.g., server room > 35˚C) or freezing conditions (e.g., pipe freeze risk < 5˚C).

BMS Action: Triggers alarms, activates backup cooling/heating, or shuts down equipment to prevent damage; escalates to on-call staff.

Benefit: Protects sensitive equipment from thermal damage, prevents pipe bursts, and ensures continuous operation of critical infrastructure.



- SX1276 wireless communication module
- 2x1.5V AAA alkaline batteries power supply
- LoRaWAN™ Class A compatible
- Frequency Hopping Spread Spectrum (FHSS)

**This is an auxiliary smart smoke sensor that must be installed alongside a fire-fighting smoke sensor; it cannot replace the fire-fighting smoke sensor.

Smoke Detection Sensor (ceiling-mounted in corridors, server rooms, electrical rooms, high-risk areas)

Use Case: Early detection of smoke from electrical faults, overheating, or fire.

BMS Action: Immediate fire alarm activation, HVAC shutdown (to prevent smoke spread), elevator recall, notification to fire response team, and integration with fire suppression systems.

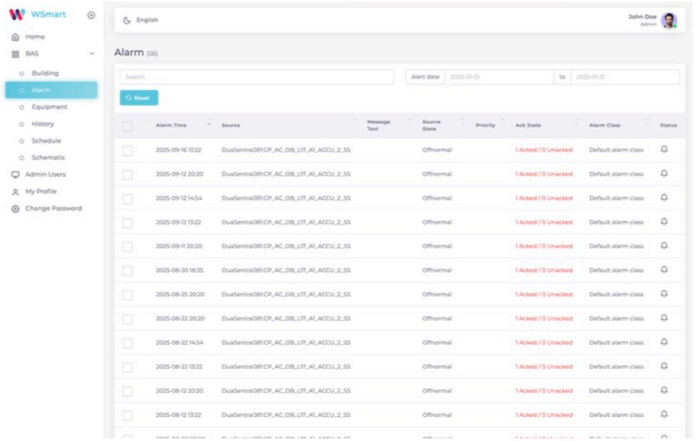
Benefit: Enhances life safety, reduces fire spread, supports compliance with fire regulations, and minimizes evacuation time and property loss.



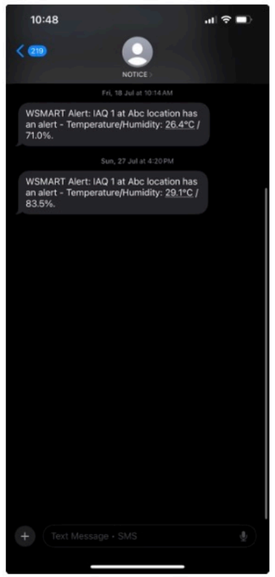
Alarms, Events & Notifications

The Alarms, Events & Notifications module ensures timely awareness and rapid response to critical and operational situations across the building.
The BMS continuously logs all system events, alarms, and user actions in a detailed event log for traceability and analysis. When an alarm or important event occurs, notifications are instantly delivered through multiple channels

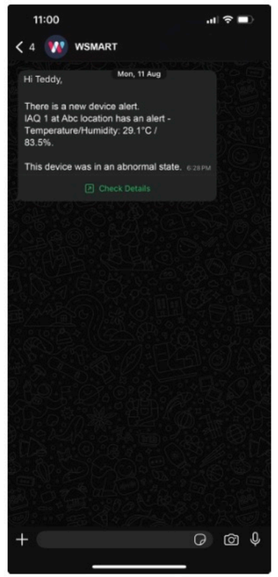
Alert Handling and Alarm log



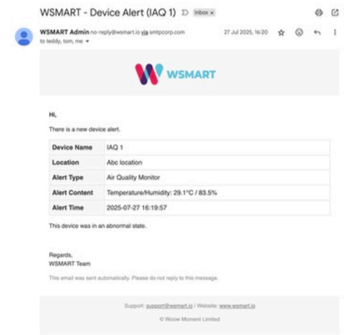
SMS



WhatsApp

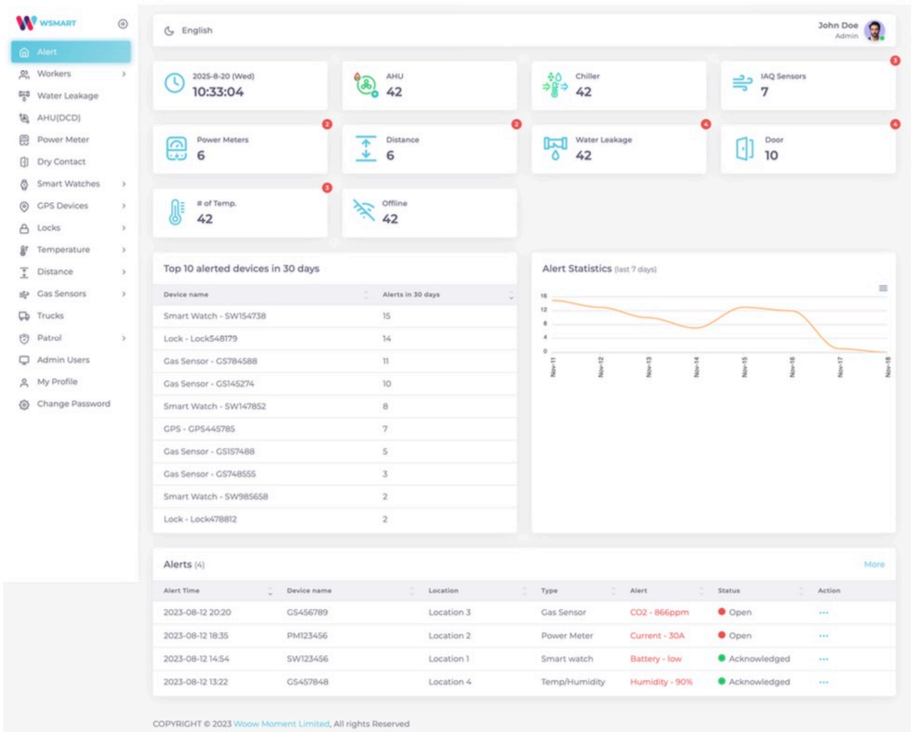


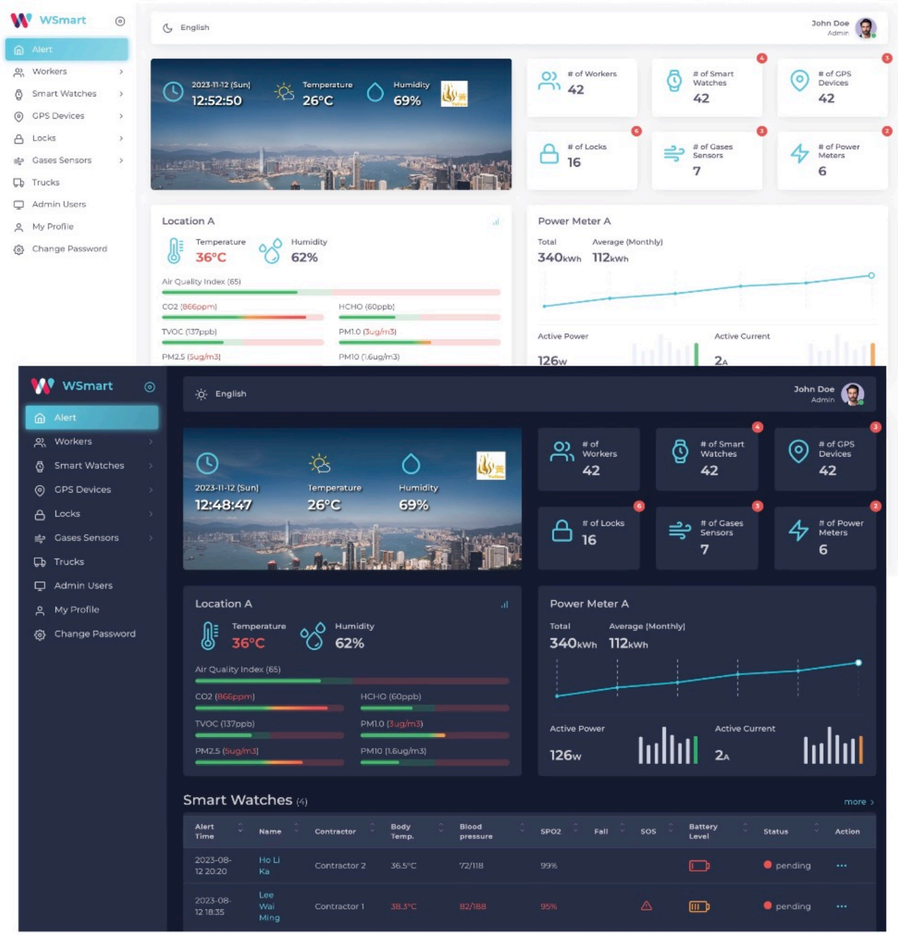
EMAIL



WSmart Platform UI

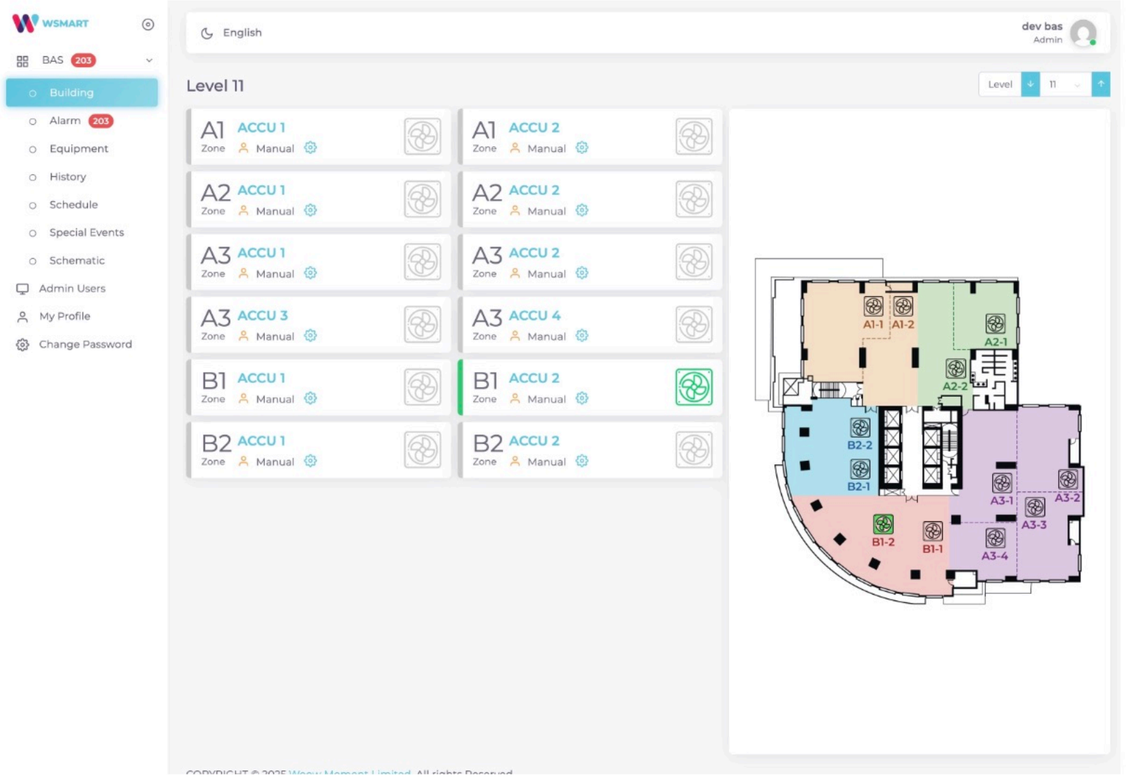
Dashboard (Customizable)

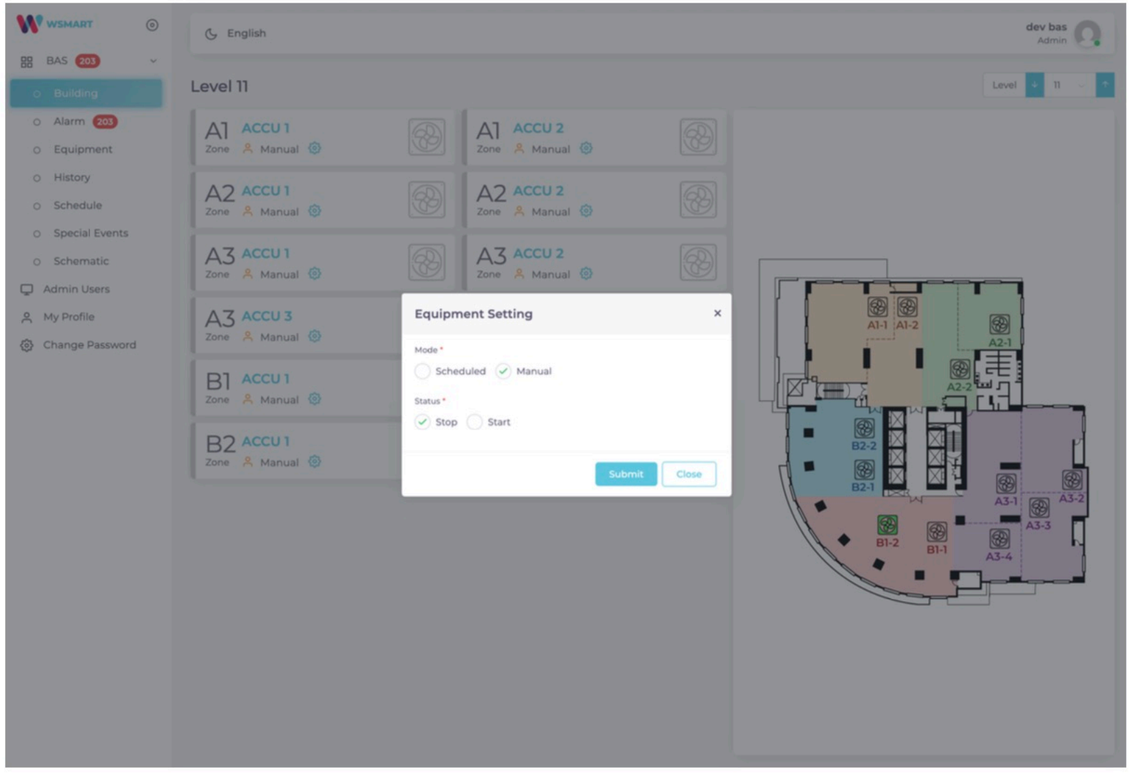




WSmart Platform UI

AHU / ACCU control

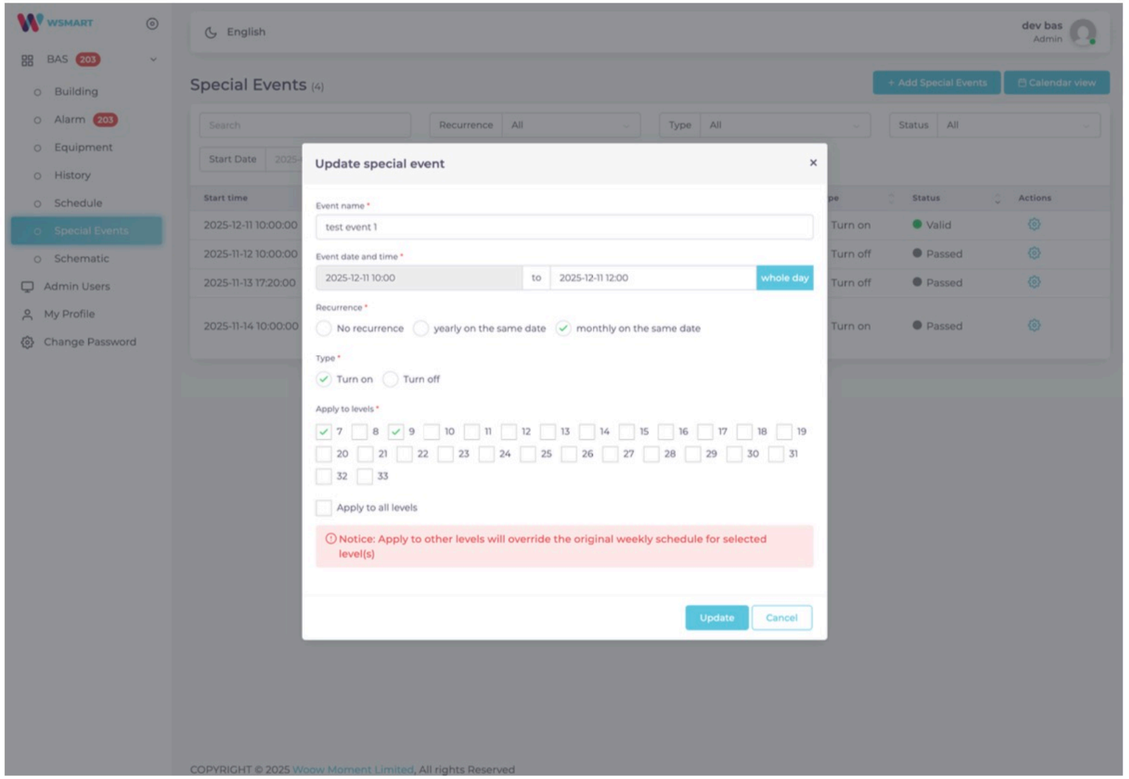


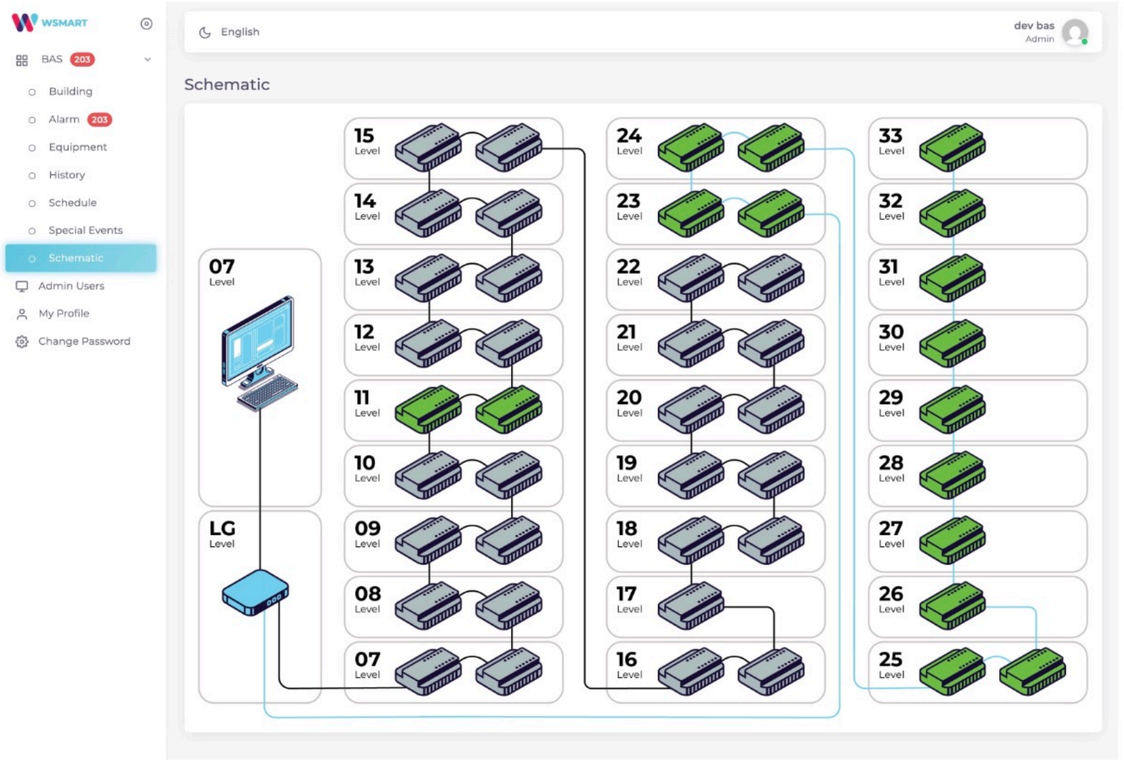


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WSmart Platform UI

Special event management and Schematic Diagram

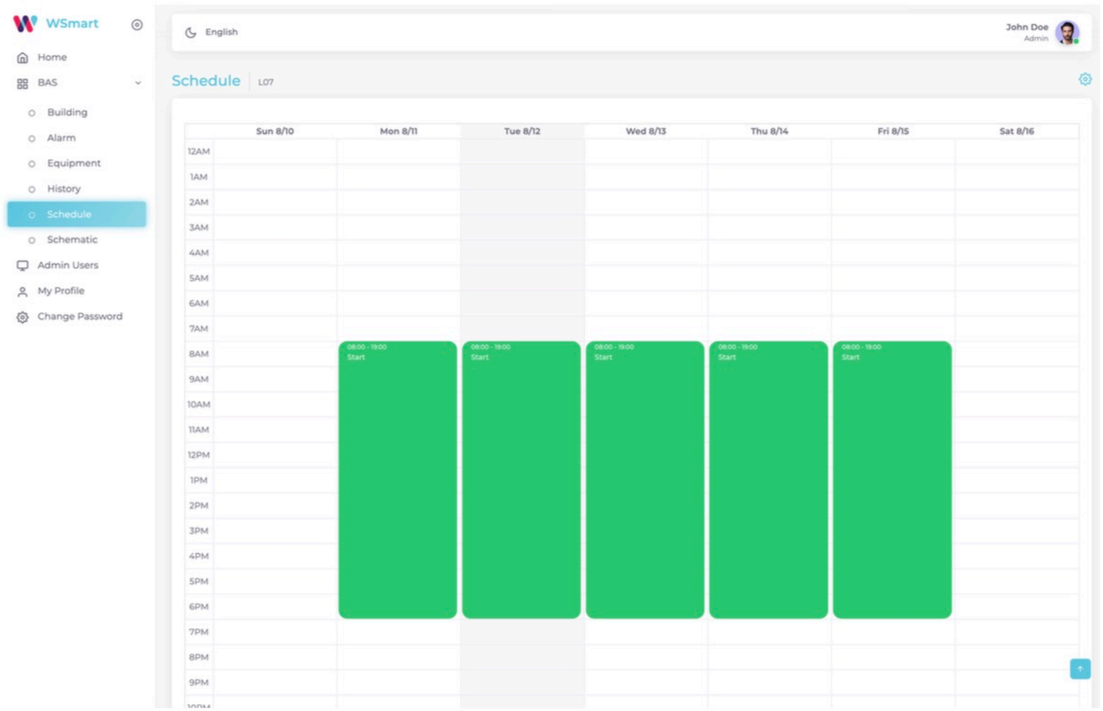


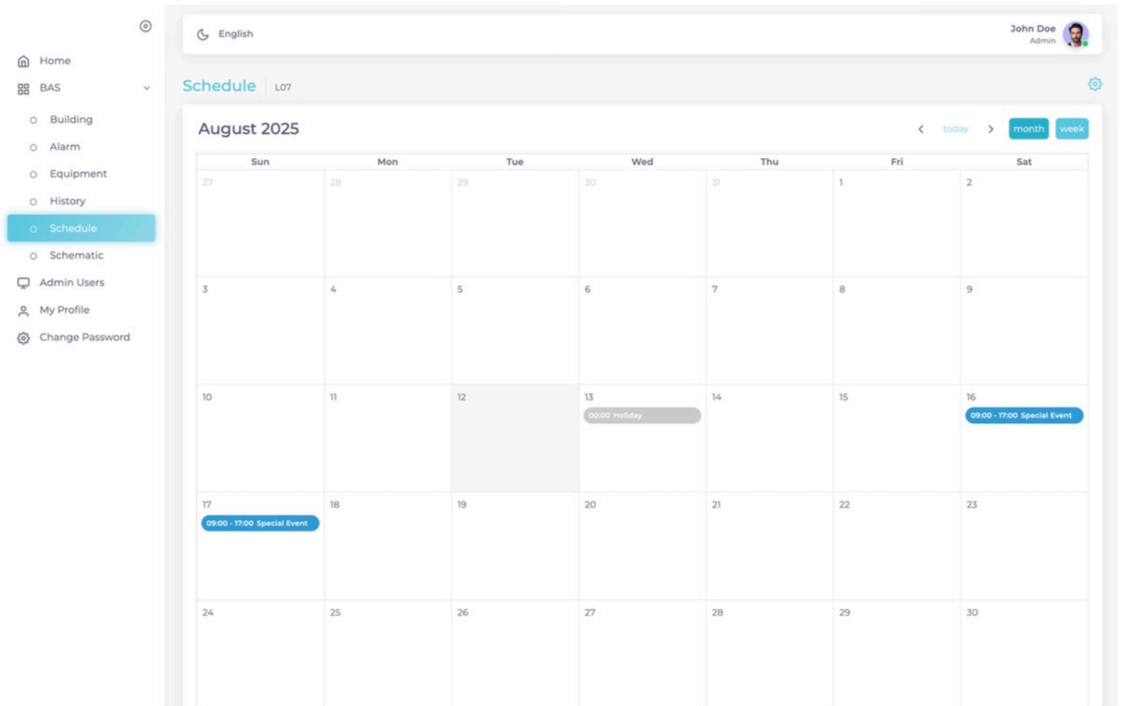


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WSmart Cloud Platform UI

Scheduling (BMS)

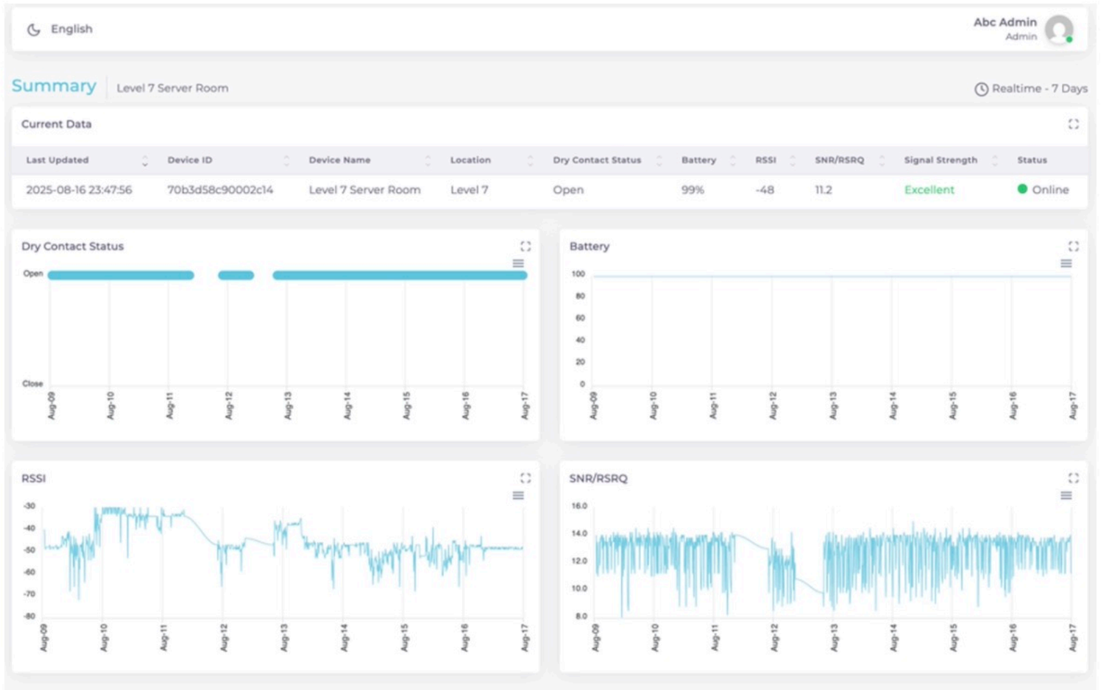


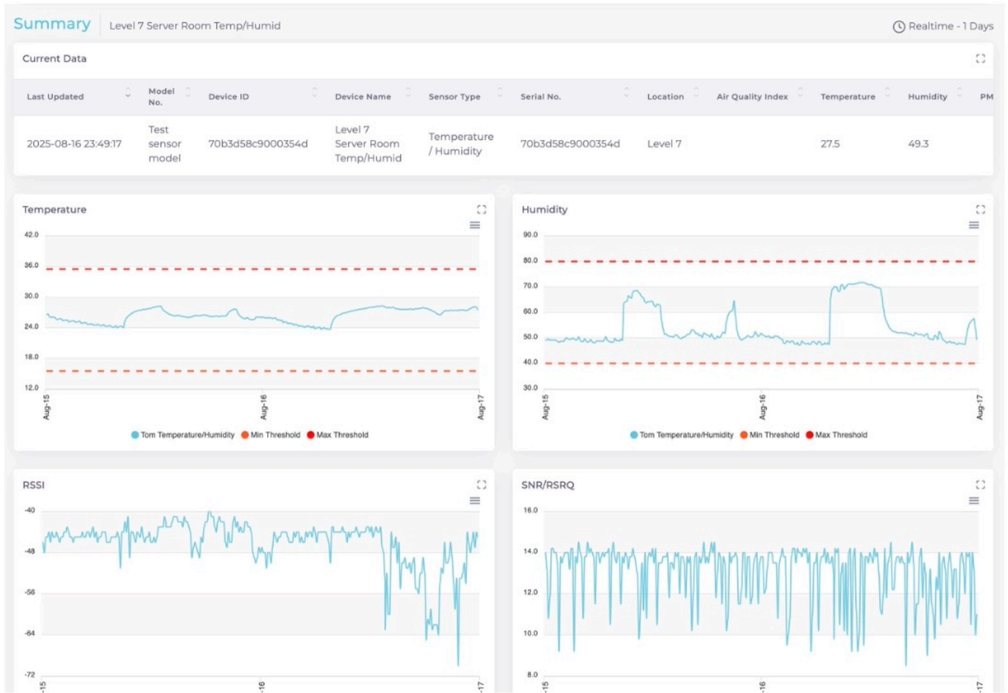


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WSmart Cloud Platform UI

Statistics and Report





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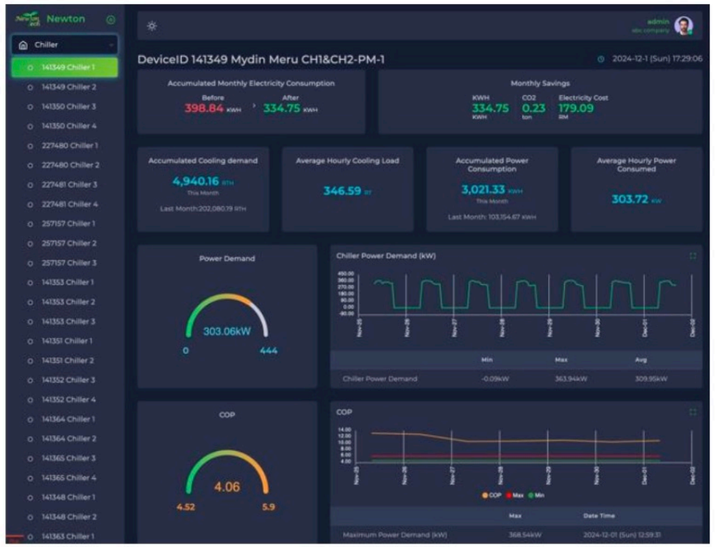
Partnering with Energy Saving service providers. We are grateful to integrate with WSMART Platform.





Innovative Water Treatment Technology

- Partner with Newton Technology Group
- We provides real-time cloud reporting, analysis, and display for the company's environmentally friendly smart system, allowing their clients to instantly understand energy-saving performance.
- We utilize IoT technology to acquire data from their energy-saving systems.





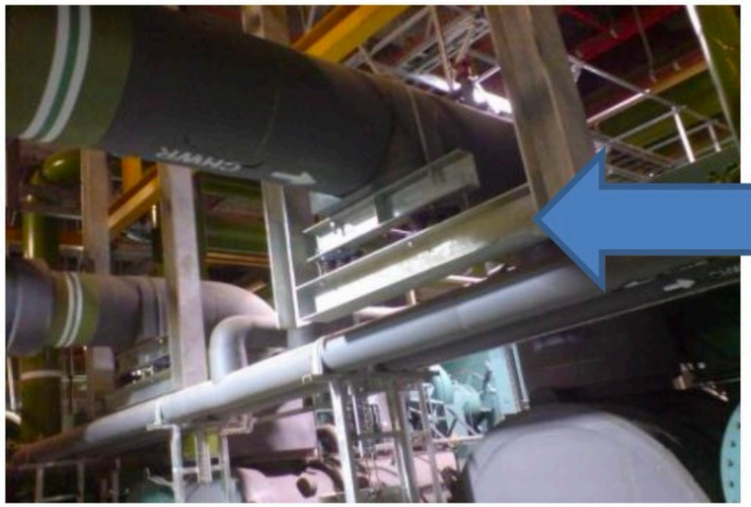
S & B Design Limited



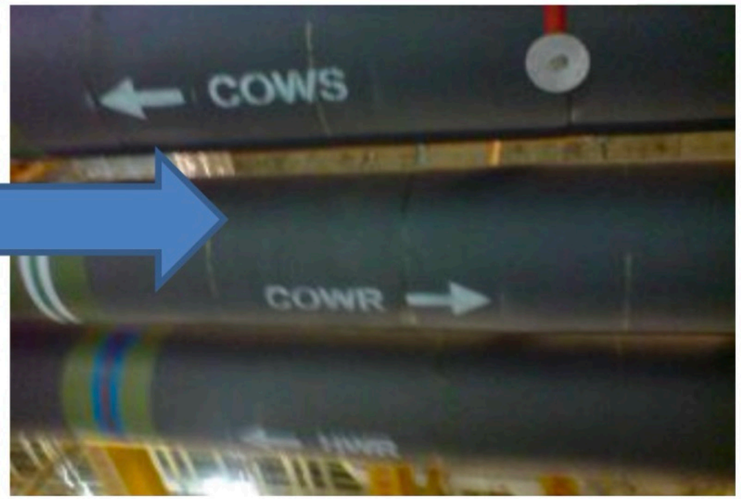
Condensing water System











Chilled water System

Patented Intelligent Eco-Friendly Energy Saving Treatment System, Consisting of below two sub-systems.

	Installation Position	Function
PS system **Chilled & condensing water system**	1. Return and supply piping of condenser. 2. Return and supply piping of heat exchanger /cooling tower 3. Return piping of Boiler 4. Supply piping of turbine	1. Applicable to all variety of water; can apply for open loop condensing and closed loop chilled water system. 2. Purifies the water by “QS Energy” generated through the transducer. 3. Descaling and prevention of scaling 4. Rust removal and prevention of rust 5. Formation of protection layer 6. Prolonging lifespan 7. Improves the flow rate





	Installation Position	Function
WS system **Integrated water treatment**	Located at cooling tower	1. Intelligent water quality monitoring system with data processing system. 2. Automatic back wash, purification, blow down system 3. Killing of bacteria, algae, legionella, microorganism. 4. Removal and prevention of scale and rust .





Key Monitoring Parameters:

Water Flow: Track the flow rate of water in the chiller system.
Chiller Water Temperatures:
Inlet Temperature
Outlet Temperature
Condensed Water Temperatures:
Inlet Temperature
Outlet Temperature
Power Consumption: Monitor the energy usage of the chiller system.

Clients Utilizing the Monitoring System
Mydin
Mines
Loh Guan Lye Hospital
Persada Convention Centre
Universiti Tenaga Nasional (Uniten)
Coherent

Technical Specifications
Cloud-based System, Web-view, Mobile Friendly, Protocol (Modbus, MQTT), LoRaWan, Linux



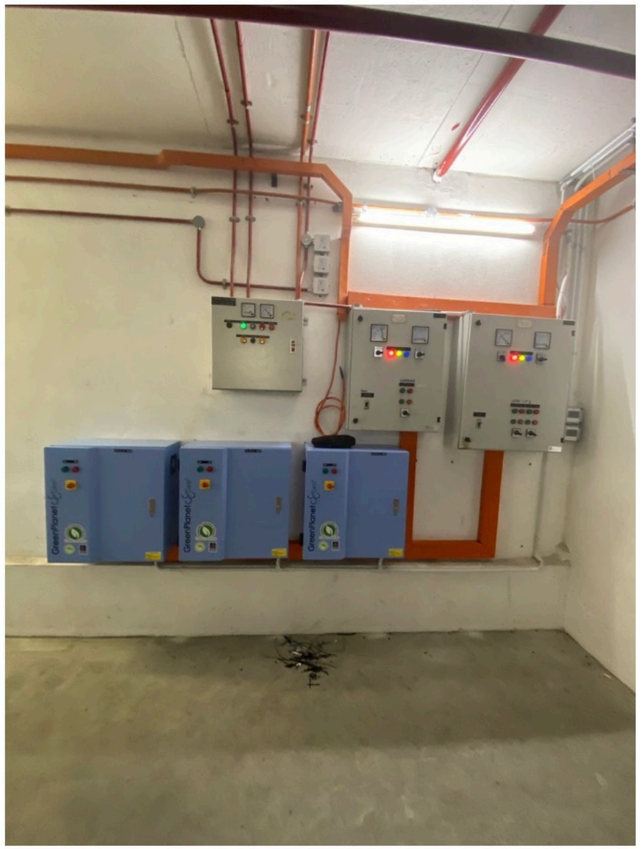
Demand Control Drives (DCD)



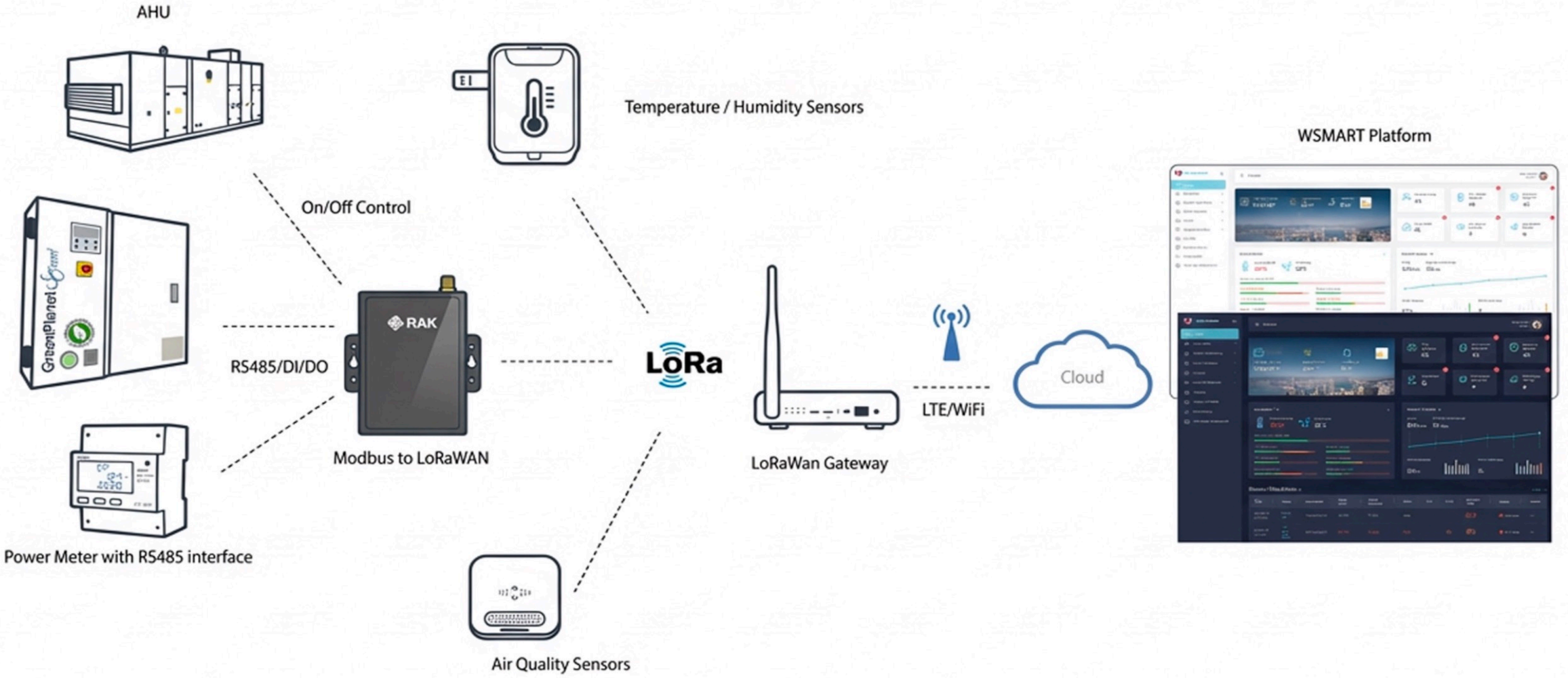
Demand Control Drives (DCD) for HVAC system and pump

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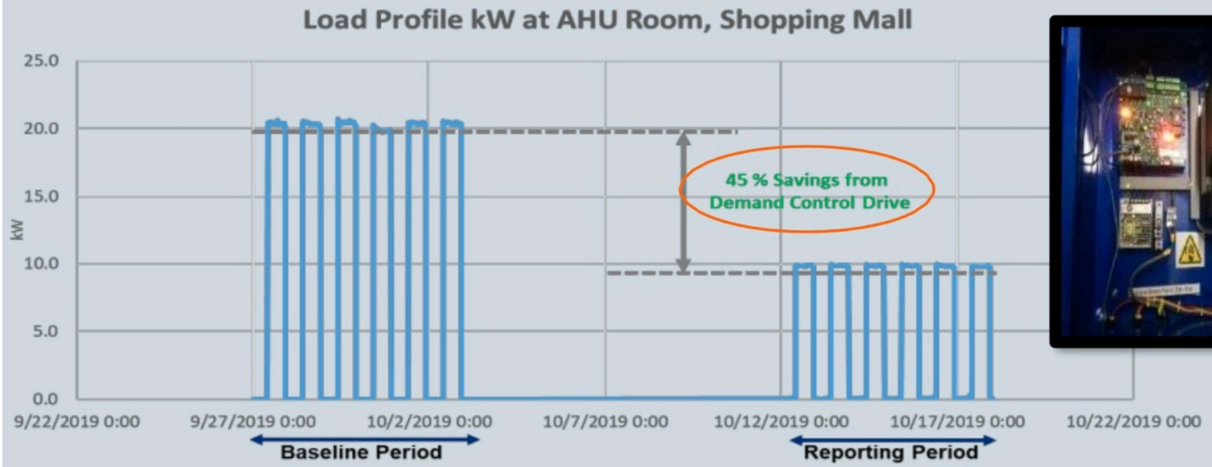
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Demand Control Drives that use computer and sensors to make systems more efficient using live data and proprietary algorithms.

1. Demand Control Drive (DCD)

BACKGROUND :

- DCD works by tracking real-time system demand and built- in algorithms automatically calculate the most efficient sequencing to HVAC
- DCD assist on manage speed and the flow thereby eliminating excessive resources
- The controlled voltages and currents to the motor result in motors running cooler and quitter saving money on maintenance
- DCD system will eliminate Inrush Currents during the start of the air handling system which can be 3 times of the running load,





30% - 45% Savings

Key Features

Monitoring Capabilities:
Power Consumption: Track and analyze power usage of the AHU.
Energy Savings: Measure and report on energy-saving metrics.
Temperature and Humidity: Monitor environmental conditions to ensure optimal performance.

Control Functionality:
AHU On/Off Control: Automate the operation of the AHU based on demand and environmental conditions.

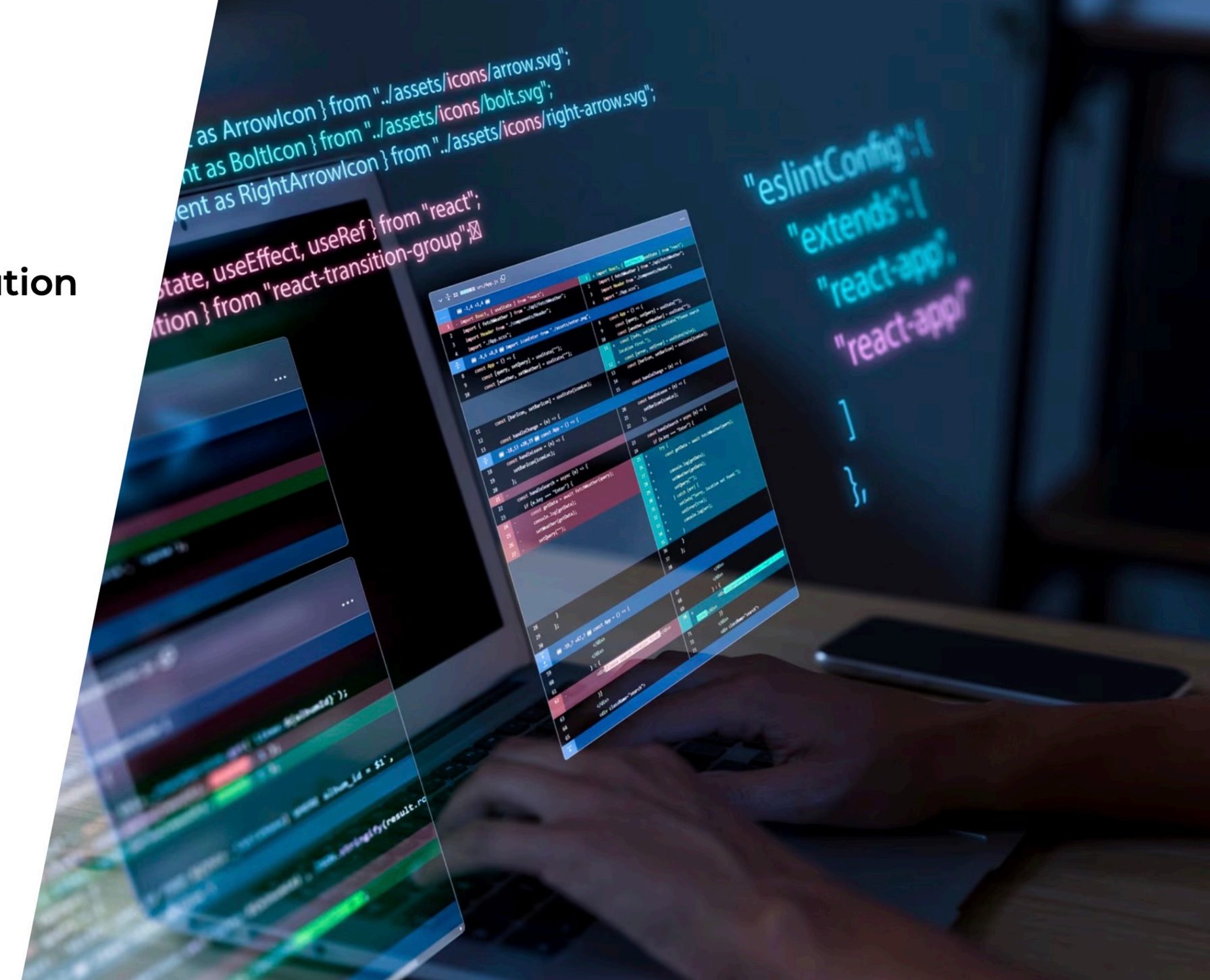
Clients Utilizing the Monitoring and Control System
Aman Sentral 88
TDK
Mydin Parit Buntar
Universiti Tenaga Nasional (Uniten)
Tenaga Nasional Berhad (TNB)

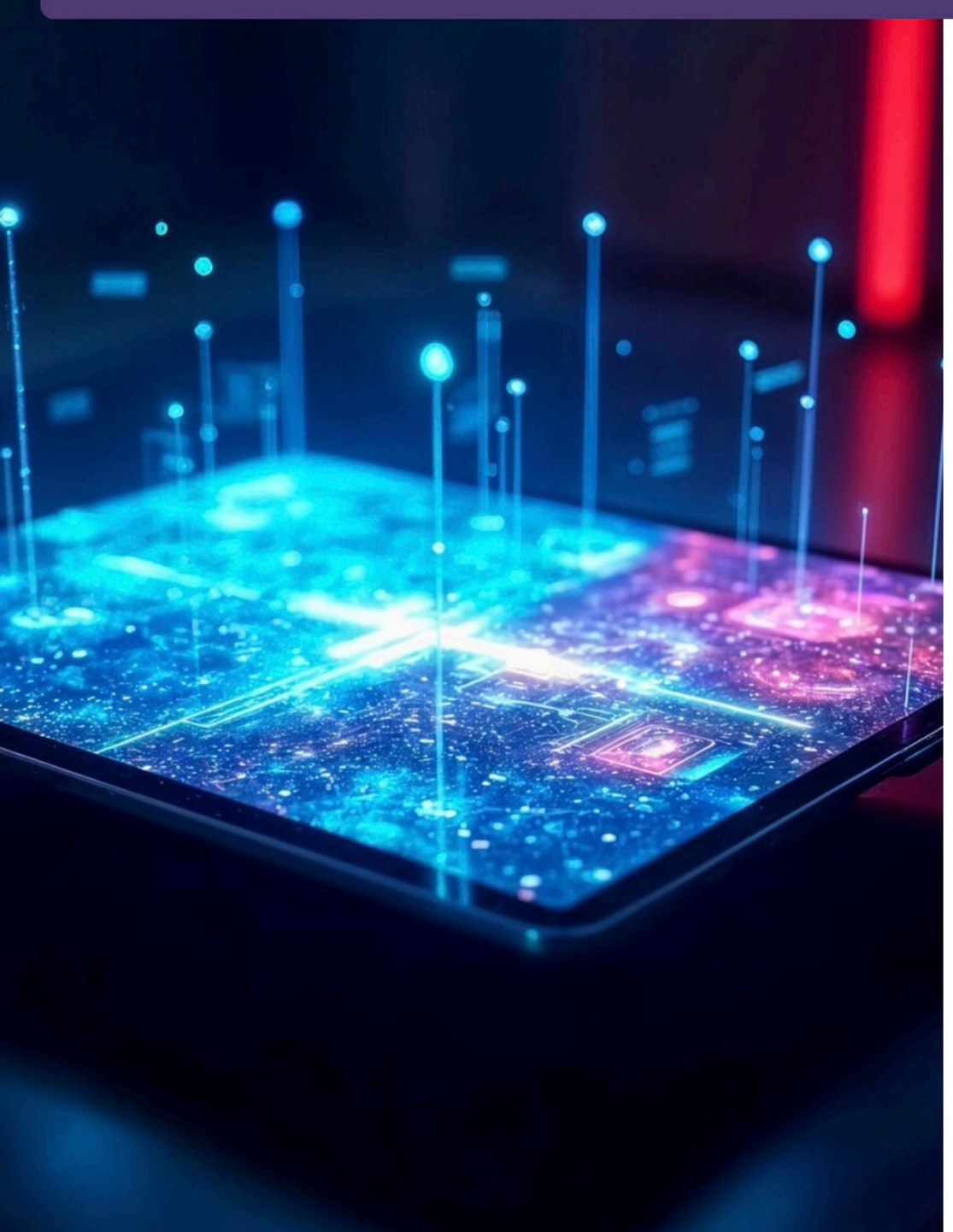
Technical Specifications
Cloud-based System, Web-view, Mobile Friendly, Protocol (Modbus, MQTT), LoRaWan, Linux

Need a customized solution for your industry?

As experienced software developers, we tailor our platform to meet your specific requirements.

Contact us to discuss your needs.





Contact Us

S & B Design Limited

email: samuelng@sandb.hk

Address: 10/F room 1001, Bonham Center, 79-85 Bonham Strand, Sheung Wan, Hong Kong